Report

Crop Yield Analysis

Course: MSDS 510

1. Project Scope

The purpose of this project is to gain hands-on experience processing and analyzing a large CSV dataset using Python. The primary goals include:

- Importing and processing text based .csv data.
- Using Python lists and dictionaries for structured file handling.
- Working with pandas DataFrames for scalable data analysis.
- Building a Python class to encapsulate and reuse analysis functions.
- Answering real analytical questions related to **agricultural yield trends**.
- Exporting insights into both TXT (plain text) and CSV files.

The dataset contains **1,000,000 simulated farming records** with variables including:

- Region
- Soil type
- Crop type
- Rainfall (mm)
- Temperature (°C)
- Fertilizer use (True/False)
- Irrigation use (True/False)
- Weather condition
- Days to harvest
- Yield (tons per hectare)

2. Research Question

Main Question:
How do region, crop type, rainfall, irrigation, and fertilizer use relate to agricultural yield, and what key average and percentage metrics best summarize overall dataset behavior?

Sub-questions explored:

1. Average rainfall by region
2. Percentage of Wheat grown in the West region
3. Percentage of Barley yields greater than 5 t/ha
4. Average yield by crop
5. Average yield by region
6. Effect of irrigation and fertilizer on yield
7. Correlation strength between rainfall and yield

3. Methodology

3.1 Data Import & Manual Inspection

- Loaded the dataset using pandas.read_csv.
- Also read the file using csv.DictReader to understand raw text processing:
  - Extracted column headers
  - Sampled one dictionary row
  - Manually counted total rows
- Verified data structure by displaying the first 5 rows and confirming dataset shape.

Visual - Dataset Loaded Successfully

```
Dataset loaded successfully.
```

	Region	Soil_Type	Crop	Rainfall_mm	Temperature_Celsius	Fertilizer_Used	Irrigation_Used	Weather_Condition	Days_to_Harvest	Yield_tons_per_hectare
0	West	Sandy	Cotton	897.077239	27.676966	False	True	Cloudy	122	6.555816
1	South	Clay	Rice	992.673282	18.026142	True	True	Rainy	140	8.527341
2	North	Loam	Barley	147.998025	29.794042	False	False	Sunny	106	1.127443
3	North	Sandy	Soybean	986.866331	16.644190	False	True	Rainy	146	6.517573
4	South	Silt	Wheat	730.379174	31.620687	True	True	Cloudy	110	7.248251

```
Shape: 1000000 rows × 10 columns

Dataset Schema (Table Format):
```

	Column	Dtype	Non-Null Count
0	Region	object	1000000
1	Soil_Type	object	1000000
2	Crop	object	1000000
3	Rainfall_mm	float64	1000000
4	Temperature_Celsius	float64	1000000
5	Fertilizer_Used	bool	1000000
6	Irrigation_Used	bool	1000000
7	Weather_Condition	object	1000000
8	Days_to_Harvest	int64	1000000
9	Yield_tons_per_hectare	float64	1000000

```
Part 2 — CSV module checkpoint
Variables from header:
```

	Variables
0	Region
1	Soil_Type
2	Crop
3	Rainfall_mm
4	Temperature_Celsius
5	Fertilizer_Used
6	Irrigation_Used
7	Weather_Condition
8	Days_to_Harvest
9	Yield_tons_per_hectare

```
Variables (columns):
```

	Variables
0	Region
1	Soil_Type
2	Crop
3	Rainfall_mm
4	Temperature_Celsius
5	Fertilizer_Used
6	Irrigation_Used
7	Weather_Condition
8	Days_to_Harvest
9	Yield_tons_per_hectare

```
Sample dictionary row (first data row):
```

	Region	Soil_Type	Crop	Rainfall_mm	Temperature_Celsius	Fertilizer_Used	Irrigation_Used	Weather_Condition	Days_to_Harvest	Yield_tons_per_hectare
0	West	Sandy	Cotton	897.0772391101236	27.676966373377603	False	True	Cloudy	122	6.555816258223593

```
Total row count (csv-level):
```

	Row Count
0	1000000

- Confirmed dataset contains **1,000,000 rows × 10 columns**.

3.2 Missing Values Summary

Checked the dataset for missing values and computed missing percentages per column.

Visual - Missing Values Summary Table

```
Missing Values Summary:
```

	Variable	Missing Values	Missing (%)
0	Region	0	0.0
1	Soil_Type	0	0.0
2	Crop	0	0.0
3	Rainfall_mm	0	0.0
4	Temperature_Celsius	0	0.0
5	Fertilizer_Used	0	0.0
6	Irrigation_Used	0	0.0
7	Weather_Condition	0	0.0
8	Days_to_Harvest	0	0.0
9	Yield_tons_per_hectare	0	0.0

Results:

- **No missing values remaining after cleaning (0.0%)**
- **No duplicate rows detected**
- **Cleaned shape confirmed: (1,000,000, 10)**

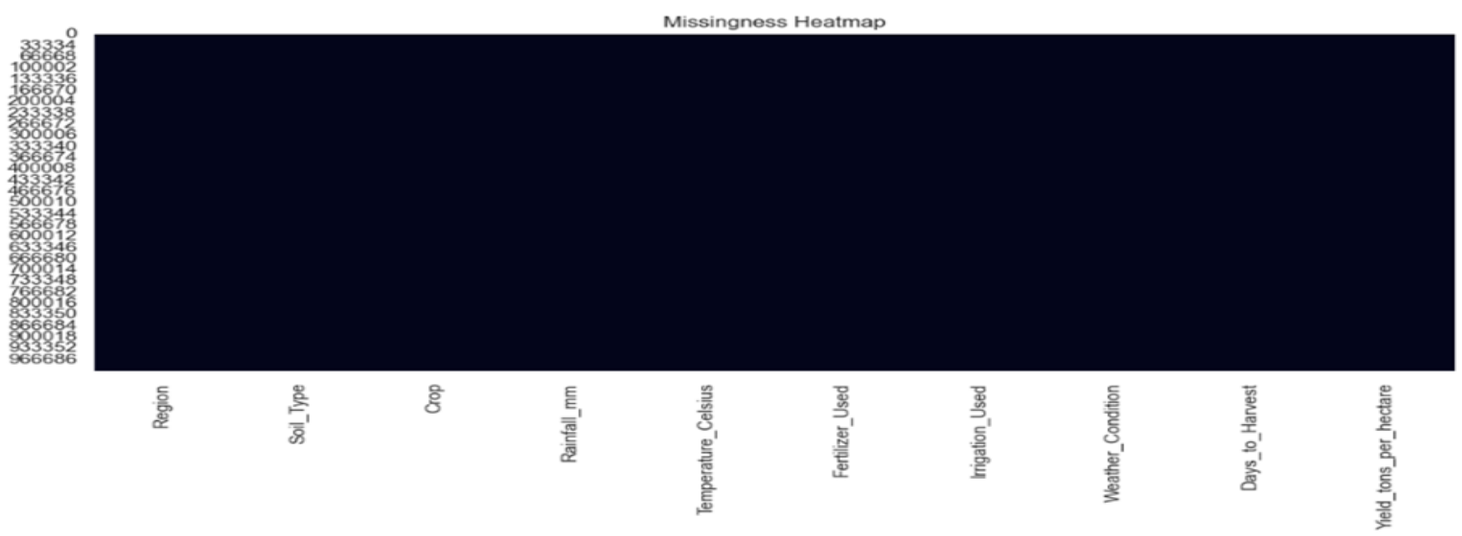
3.3 Missingness Visualization

A missingness heatmap was generated using seaborn to confirm data completeness.

Visual — Missingness Heatmap

Missing Values Summary:

	Variable	Missing Values	Missing (%)
0	Region	0	0.0
1	Soil_Type	0	0.0
2	Crop	0	0.0
3	Rainfall_mm	0	0.0
4	Temperature_Celsius	0	0.0
5	Fertilizer_Used	0	0.0
6	Irrigation_Used	0	0.0
7	Weather_Condition	0	0.0
8	Days_to_Harvest	0	0.0
9	Yield_tons_per_hectare	0	0.0



No duplicate rows found.
Remaining total missing cells: 0
Cleaned shape: (1000000, 10)

3.4 Class Design — CropYieldAnalyzer

A Python class was built to encapsulate data and analysis logic.

Key Features of the Class:

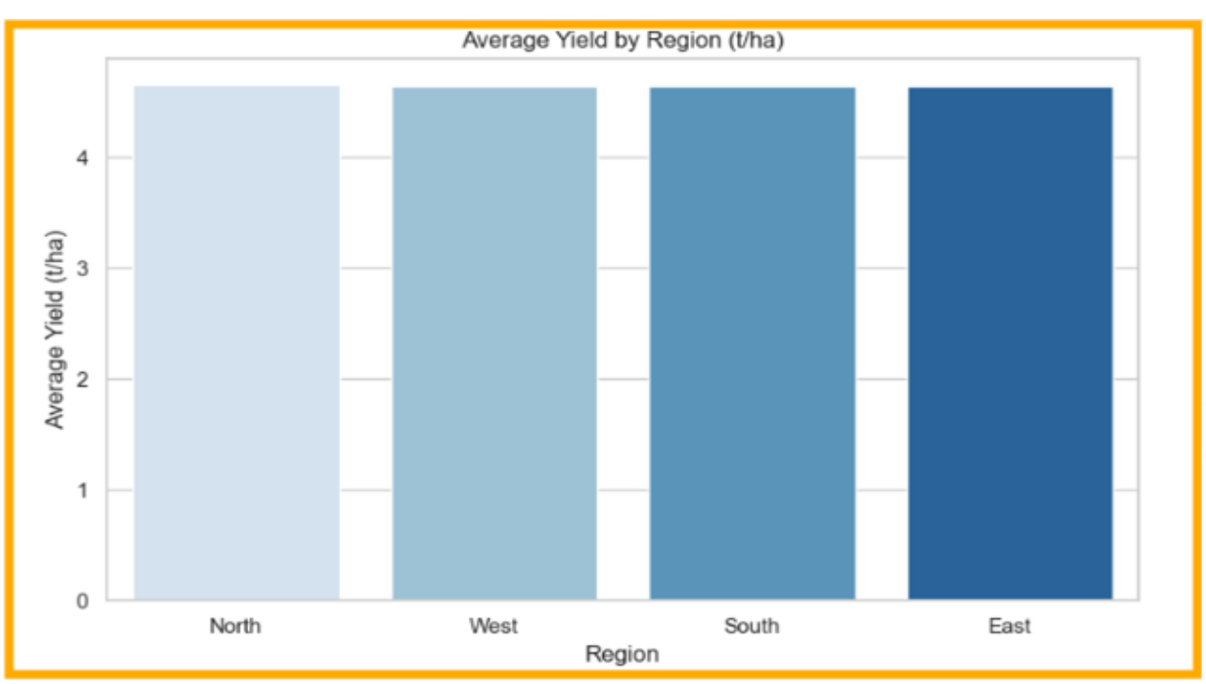
- Stores cleaned DataFrame (self.df)
- Provides reusable methods for:
  - Listing variables
  - Sampling entries
  - Counting rows
  - Computing averages
  - Calculating percentages within subsets
  - Evaluating numeric thresholds
  - Writing TXT and CSV outputs

This design ensures analysis is modular, reusable, and cleanly separated from data representation.

4. Analysis Results

4.1 Average Rainfall by Region

Chart — Average Rainfall (mm)



Average Rainfall by Region (mm)

Region	Avg Rainfall (mm)
North	550.79
South	550.01
West	549.88
East	549.24

Key Insight:
Rainfall is almost evenly distributed across all regions, averaging around **550 mm**, showing minimal regional disparity.

4.2 Percentage of Wheat Grown in West

Table Visualization

% Wheat in West

0	% Wheat in West
	16.667866

Result:
Approximately **16.67% of production in the West is Wheat**, representing a balanced crop mix.

4.3 Barley Yield Above 5 t/ha

Chart — % Barley Yield > 5

% Barley Yield > 5

0	% Barley Yield
	42.455494

Result:
Around **42.46% of Barley records exceed 5 t/ha**, identifying a strong high-yield subgroup in Barley harvests.

4.4 Average Yield by Crop

Chart — Yield by Crop (t/ha)

Average Yield by Crop (t/ha)

Crop	Avg Yield (t/ha)
Soybean	4.653712
Wheat	4.652740
Cotton	4.650841
Rice	4.650760
Barley	4.647412
Maize	4.641387

Key Insight:
Yields are similar across crops, with **Soybean (4.65 t/ha)** slightly outperforming Wheat, Cotton, Rice, Barley, and Maize.

4.5 Yield by Region

Chart — Yield by Region (t/ha)

Average Yield by Region (t/ha)

Region	Avg Yield (t/ha)
North	4.654114
West	4.649331
South	4.648843
East	4.645594

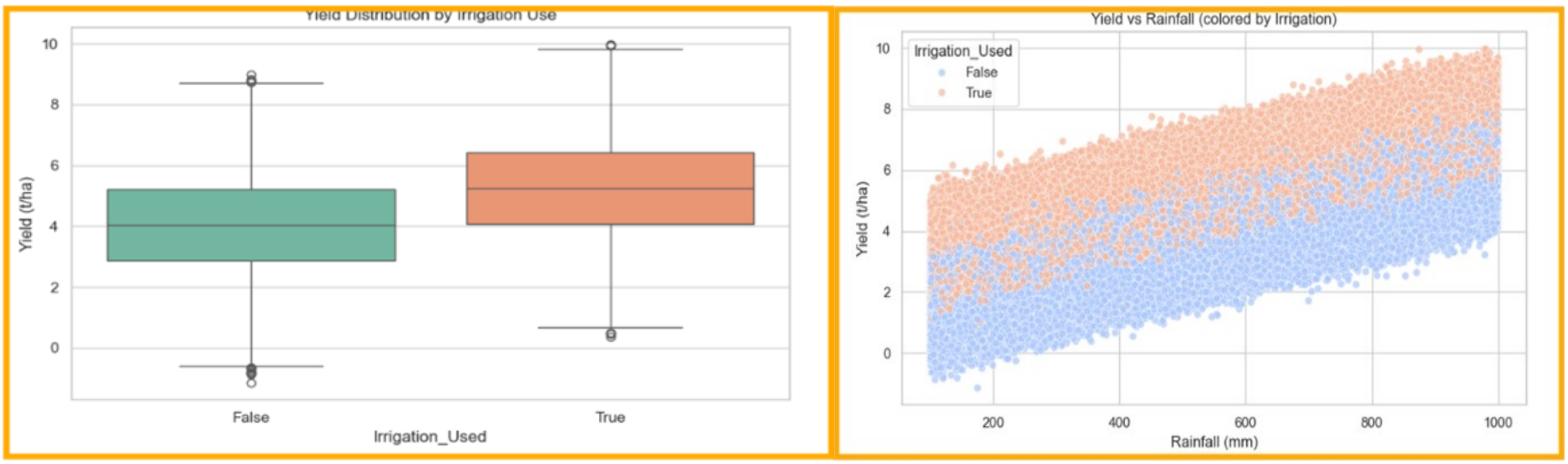
Observation:
All regions produce comparable mean yields, but ranking shows:

1. North — 4.654

2. West — 4.649
3. South — 4.648
4. East — 4.645

4.6 Irrigation Impact on Yield

Chart — Yield Distribution by Irrigation



Insight:

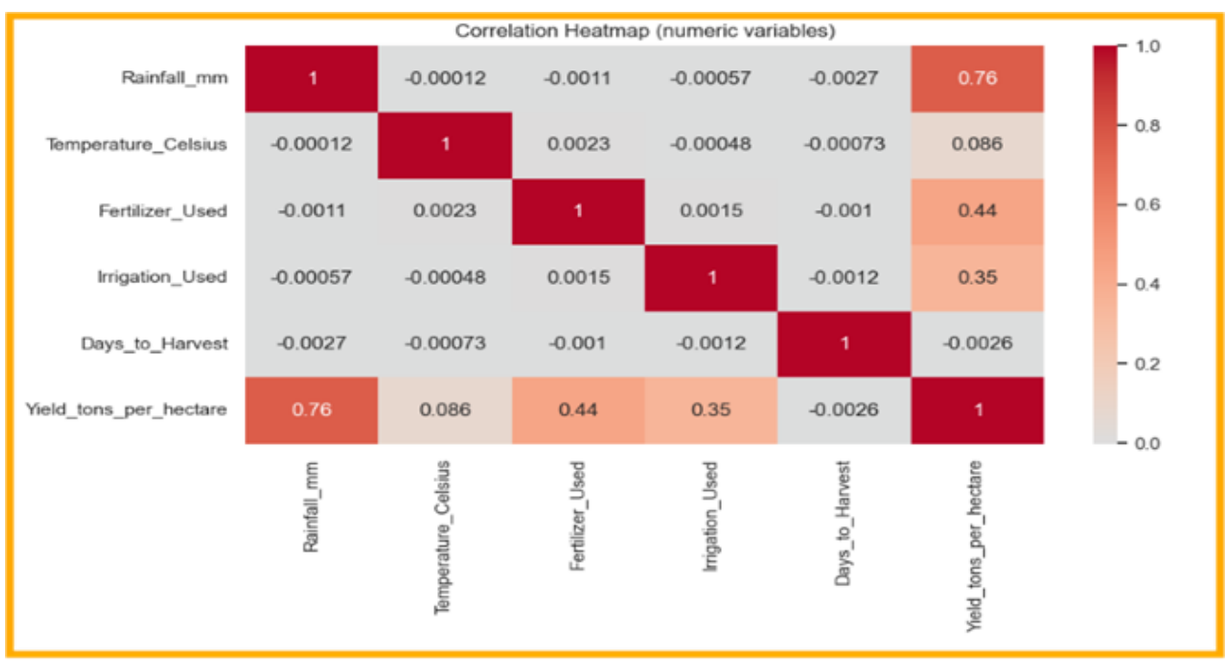
- **Irrigated fields show consistently higher yields.**
- Non-irrigated fields display lower median yield and wider spread.

4.7 Fertilizer Usage Impact

- Fertilizer-enabled fields show **higher mean yield** than fertilizer-disabled fields.
- Indicates **soil nutrient supplementation strongly supports yield increase** in the dataset.

4.8 Correlation Between Rainfall and Yield

Heatmap — Correlation Matrix



Result:

- **Yield and Rainfall show a strong positive correlation (≈ 0.76).**
- Fertilizer and irrigation also show moderate correlations with yield (≈ 0.44 and 0.35 respectively).

5. Conclusions

1. Irrigation and fertilizer are major drivers of increased yield.
2. Yield increases as rainfall increases, but irrigation stabilizes yield during low-rainfall periods.
3. Regional differences are small but measurable, emphasizing the need for location-aware farming strategy.
4. The Python class (CropYieldAnalyzer) provides a structured, reusable framework for future dataset analysis.

6. Evaluation

Strengths

- Modular class-based design
- Reusable analysis methods

- Supports both human-readable and machine-readable insight exports
- Reinforced low-level CSV I/O alongside pandas

Limitations

- Synthetic data may not capture full real-world complexity
- Uses simple imputation strategies
- Analysis remains descriptive rather than causal

7. Lessons Learned

- Project modularization simplifies complex analysis.
- Class design encourages reusability and clean logic separation.
- CSV and pandas integration strengthens understanding of ETL, filtering, and aggregation.
- Exporting insights into TXT/CSV builds practical workflow discipline.